Project proposal feedback

Project title: Music Recommendation with Knowledge-Graph-Based Prediction
Group: Pedro Fanica, Aarthi Perumpillichira, Maja Skwaron

Overall assessment

This feedback is based on the pivot to music recommendation.

What is already good

- The pivot removes the excessive scope of the original transcription → harmony → graph → recommendation pipeline.
- The project is now more feasible within the course timeline, while still having some novelty.

Main issues to address

- The task still needs to be defined very precisely:
  - predict whether a user will like a song?
  - rank candidate songs for a user?
  - next-song recommendation?
- The proposal must explain **how the knowledge graph enters the model**:
  - graph embeddings?
  - node features?
  - path-based similarity?
  - graph neural network?
- The group should avoid overloading the project again by trying too many components at once.
- The baseline design becomes even more important now: you must compare the graph-based approach against simpler recommenders.

Detailed feedback

1. Problem definition

The project should now be stated in a form like – this is just one possible formulation:

> Given user interaction history and a music knowledge graph encoding relations between songs, artists, genres, chords, or other musical entities, recommend songs that the user is likely to enjoy.

2. Data

The proposal should now describe two data sources:

- **user-item interaction data**, which defines the recommendation target
- **the music knowledge graph**, which provides structured side information

The important thing is to explain how these two are linked. For example:

- does each song in the interaction dataset correspond to a node in the graph?
- what kinds of edges exist in the graph?
- are there attributes such as genre, artist, harmonic similarity, key, instrumentation, or mood?

Without this alignment, the graph cannot actually be used effectively.

3. Method and baseline

This is now the core design question. A good project structure would be:

- **Baseline 1:** popularity-based recommender
- maybe - Baseline 2: collaborative filtering / matrix factorization
- **Main model:** recommendation model that uses graph-derived representations

Depending on ambition and feasibility, the graph-based model could be:

- matrix factorization plus graph embeddings
- neural collaborative filtering with graph features
- a graph neural network if the graph is clean and well aligned

4. Evaluation

The evaluation should now clearly use recommendation metrics such as:

- **Recall@K**
- **Precision@K**
- **NDCG@K**

These should be the main metrics. You should compare:

- interaction-only baselines
- graph-enhanced model

That comparison is the real scientific question: **does the knowledge graph help recommendation?**

5. Feasibility

This version is **feasible**, provided the graph already exists or can be reused from the other course without major extra effort. The main risks are:

- alignment between graph entities and recommendation items
- spending too much time polishing the graph instead of running recommendation experiments

The safest version is:

- fixed interaction dataset
- fixed graph
- one simple recommender baseline
- one graph-enhanced recommender

6. Experimental depth

This project now has a good opportunity for real depth. A strong question would be:

- Does graph-based side information improve recommendation over collaborative filtering alone?
- Which graph relations are most useful for recommendation?
- Does the graph help more in sparse-data or cold-start settings?

Any one of these would make the project stronger than a generic recommender.

How to improve the proposal

- Define one precise recommendation task.
- Explain exactly how the graph is connected to the recommendation model.
- Use simple recommendation baselines first.
- Compare **without graph** vs **with graph**.

Recommended version of the project

> We study music recommendation from user-item interaction data enriched with a music knowledge graph built in a parallel course. In this course, the graph is used as structured side information to improve recommendation quality. We compare standard recommender baselines, such as popularity and collaborative filtering, against a graph-enhanced model that incorporates musical relations between songs, artists, genres, and related entities. The main goal is to evaluate whether graph-based musical knowledge improves top-(K) recommendation quality.

Suggested minimum plan for success

By **Milestone 1**, the group should already have:

- fixed the interaction dataset
- fixed the subset of the knowledge graph to be used
- aligned songs/items between the interaction data and the graph
- implemented one simple recommendation baseline
- defined the evaluation split and ranking metrics
- chosen the graph-based recommendation model

By the **final report**, a solid project should include:

- one clearly defined recommendation task
- one or two standard baselines
- one graph-enhanced model
- Recall@K / Precision@K / NDCG@K results
- analysis of when the graph helps
- discussion of limitations, especially graph coverage and item alignment

Final recommendation

- approved with minor changes.